

Physics Mock Test

Electric Charges and Fields

Instructions: This mock test covers key concepts including Coulomb's law, electric fields, Gauss's law, and related phenomena.

Total Questions: 12 (6 MCQs, 6 Numericals) | **Time Suggested:** 60 minutes.

Section 1: Multiple Choice Questions (1 mark each)

1. Coulomb's Law

Two point charges, $+q$ and $-q$, are separated by a distance r . The force between them is:

- a) Attractive and proportional to $1/r^2$
- b) Repulsive and proportional to $1/r$
- c) Attractive and proportional to r^2
- d) Repulsive and proportional to $1/r^2$

2. Electric Field

The electric field due to a point charge Q at a distance r is $E = kQ/r^2$. If the charge is doubled and the distance is halved, the new electric field is:

- a) 4 times the original
- b) 8 times the original
- c) 2 times the original
- d) 16 times the original

3. Gauss's Law

Gauss's law states that the total electric flux through a closed surface is equal to:

- a) The charge enclosed divided by ϵ_0
- b) The charge enclosed multiplied by ϵ_0
- c) The charge enclosed divided by $4\pi\epsilon_0$
- d) The charge enclosed multiplied by $4\pi\epsilon_0$

4. Conductors and Fields

Inside a hollow conductor with excess charge on its surface, the electric field is:

- a) Zero
- b) Equal to the field outside
- c) Twice the field outside
- d) Half the field outside

5. Dipole Moment

The electric field due to an electric dipole at a point along its axis is proportional to:

- a) $1/r^2$
- b) $1/r^3$
- c) $1/r$
- d) r^2

6. Superposition Principle

The principle of superposition for electric fields means that:

- a) Fields add vectorially
- b) Fields add algebraically
- c) Fields multiply
- d) Fields cancel out completely

Section 2: Numerical/Short-Answer Problems (2-3 marks each)

7. Coulomb's Law Calculation

Two charges, $+4 \mu\text{C}$ and $-2 \mu\text{C}$, are placed 10 cm apart in vacuum. Calculate the magnitude of the force between them. (Use $k = 9 \times 10^9 \text{ N m}^2/\text{C}^2$)

8. Electric Field Due to a Charge

A point charge of $+5 \text{ nC}$ is placed at the origin. Find the electric field at a point (3 m, 4 m) in the xy-plane.

9. Gauss's Law Application

A spherical conductor of radius 5 cm has a charge of $+10\text{ }\mu\text{C}$ on its surface. Using Gauss's law, find the electric field at a point 3 cm from the center (inside the conductor).

10. Electric Dipole

An electric dipole consists of charges $+q$ and $-q$ separated by distance $2a$. The dipole moment $p = 2qa$. Find the electric field at a point on the equatorial line at a distance r from the center, where $r \gg a$.

11. Potential Difference

The electric potential due to a point charge Q at a distance r is $V = kQ/r$. If a charge is moved from $r_1 = 2\text{ m}$ to $r_2 = 4\text{ m}$ from a $+5\text{ }\mu\text{C}$ charge, calculate the work done by the electric field. (Assume no other forces.)

12. Capacitance and Fields

A parallel-plate capacitor has plates of area 100 cm^2 separated by 1 mm. If the dielectric constant is 4, calculate the capacitance. ($\epsilon_0 = 8.85 \times 10^{-12}\text{ F/m}$)

Answer Key (For Self-Evaluation)

1. a) Attractive and proportional to $1/r^2$
2. b) 8 times the original ($E' = k(2Q)/(r/2)^2 = 8kQ/r^2$)
3. a) The charge enclosed divided by ϵ_0 ($\Phi = Q_{\text{encl}} / \epsilon_0$)
4. a) Zero
5. b) $1/r^3$ (For a dipole, $E \sim 1/r^3$ at large distances)
6. a) Fields add vectorially
7. $F = k|q_1q_2|/r^2 = (9 \times 10^9)(4 \times 10^{-6})(2 \times 10^{-6}) / (0.1)^2 = 7.2 \text{ N}$
8. $r = \sqrt{3^2 + 4^2} = 5 \text{ m}$. $E = (9 \times 10^9)(5 \times 10^{-9}) / 5^2 = 1.8 \text{ N/C}$ (Directed away from origin along unit vector $(0.6i + 0.8j)$)
9. $E = 0$ (Inside a conductor, the net electric field is zero as all charge resides on the surface)
10. $E = k p / r^3$ (On equatorial line, opposite to the direction of p)
11. $W = kQq (1/r_1 - 1/r_2) = (9 \times 10^9)(5 \times 10^{-6})(5 \times 10^{-6}) (0.5 - 0.25) = 0.05625 \text{ J}$
12. $C = K\epsilon_0 A/d = 4 \times (8.85 \times 10^{-12}) \times (0.01) / (0.001) = 3.54 \times 10^{-10} \text{ F}$